

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0624

Picu 138 kV
May 28, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0624. The Interconnection Customer proposes a 278.8 MW large load (data center) facility interconnecting at the Picu 138 kV substation in the Kentucky area. The request was received on June 02, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 212 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$60,869,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WELLINGTON – DAVISVILLE 138 KV	MIRABEL – WEC 138 KV Line	174.0	333	16.3
LESLIE – ST JOSEPH 138 KV L	WIXOM – ALBION ROAD 138 KV	169.1	444	14.5
NORTH APPLETON – LESLIE 138 KV	ALBON RD – WIXOM 138 KV Line	168.0	451	14.5
DAVISVILLE – N APPLETON 138 KV L	ALBION RD – WIXOM 138 KV	166.4	460	14.5
MIRABEL – DAVISVILLE 138 KV	WEC – WELLINGTON SUBSTATION 138 KV	165.2	308	5.0
MIRABEL – WEC 138 KV	WEC – WELLINGTON 138 KV	162.9	317	5.0
WEC – WELLINGTON 138 KV	MIRABEL – WEC 138 KV	148.0	451	6.1
PICU – HAMILTON WOLFE RD 138 KV line	BEAVER CRK – PICU 138 KV	137.1	211	100.0
BULAN – HAZARD 138 KV Line 1	BULAN – HAZARD 138 KV Circuit 2	133.9	374	8.5
BULAN – HAZARD 138 KV Line 2	HAZARD – BULAN 138 KV Cir. 1	133.9	374	8.5
SANTA ROSA BEACH SUBSTATION – TULE 138 KV	DAVISVILLE SUBSTATION – N APPLETON SUBSTATION 138 KV	133.8	148	5.9
BERRY ST – WISSOTA 138 KV	DAVISVILLE – N APPLETON 138 KV	133.8	421	7.5
HAZARD – HAMILTON WOLFE RD 138 KV	PREBLE – FERNWOOD 138 KV Line	132.9	256	35.8
PICU – BEAVER CREEK 138 KV	PICU – HAMILTON WOLFE ROAD 138 KV	128.7	224	100.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BEAVER CREEK – MANHATTAN 138 KV	PICU – HAMILTON WOLFE ROAD 138 KV	118.5	218	70.0
PREBLE – FERNWOOD 138 KV L	HAZARD – HAMILTON WOLFE ROAD 138 KV Line	109.3	247	17.0
WEST MOUNT VERNON – SANTA ROSA BEACH 138 KV L	DAVISVILLE – N APPLETON 138 KV	107.3	196	5.9
WISSOTA – NSP 138 KV	DAVISVILLE – N APPLETON 138 KV	106.8	223	5.9
NSP – W MOUNT VERNON SUBSTATION 138 KV	N APPLETON – DAVISVILLE 138 KV	105.4	213	5.9
FERNWOOD – CHESTNUT 138 KV L	CHESTNUT SUBSTATION – CONSOLIDATED COAL 138 KV	101.6	504	15.9
BULAN – FERNWOOD 138 KV	PREBLE SUBSTATION – FERNWOOD 138 KV L	100.3	265	11.8
CONSOLIDATED COAL – CHESTNUT 138 KV L	FERNWOOD – CHESTNUT 138 KV Line	100.1	448	12.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WELLINGTON – DAVISVILLE 138 KV	Rebuild line with higher-capacity conductor	\$2,045,000	8% (\$163,000)
LESLIE – SAINT JOSEPH 138 KV	Rebuild line with higher-capacity conductor	\$13,572,000	6% (\$756,000)
LESLIE – NORTH APPLETON 138 KV	Rebuild line with higher-capacity conductor	\$5,254,000	6% (\$293,000)
DAVISVILLE – N APPLETON 138 KV L	Rebuild line with higher-capacity conductor	\$7,000,000	6% (\$390,000)
MIRABEL – DAVISVILLE 138 KV	Rebuild line with higher-capacity conductor	\$20,601,000	4% (\$711,000)
WEC – MIRABEL 138 KV	Rebuild line with higher-capacity conductor	\$5,310,000	4% (\$183,000)
WEC – WELLINGTON 138 KV	Rebuild line with higher-capacity conductor	\$32,678,000	3% (\$1,034,000)
PICU – HAMILTON WOLFE RD 138 KV line	Rebuild line with higher-capacity conductor	\$9,963,000	100% (\$9,963,000)
BULAN – HAZARD 138 KV Line 1	Rebuild line with higher-capacity conductor	\$3,052,000	10% (\$315,000)
BULAN – HAZARD 138 KV Line 2	Rebuild line with higher-capacity conductor	\$3,007,000	10% (\$310,000)
SANTA ROSA BEACH SUBSTATION – TULE 138 KV	Rebuild line with higher-capacity conductor	\$5,854,000	8% (\$489,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BERRY ST – WISSOTA 138 KV	Rebuild line with higher-capacity conductor	\$2,067,000	4% (\$88,000)
HAMILTON WLFE RD – HAZARD 138 KV L	Rebuild line with higher-capacity conductor	\$11,808,000	100% (\$11,808,000)
PICU – BEAVER CREEK 138 KV	Rebuild line with higher-capacity conductor	\$22,180,000	100% (\$22,180,000)
BEAVER CREEK – MANHATTAN 138 KV	Rebuild line with higher-capacity conductor	\$3,683,000	100% (\$3,683,000)
FERNWOOD – PREBLE 138 KV	Reconductor line	\$5,596,000	100% (\$5,596,000)
SANTA ROSA BEACH – W MOUNT VERNON 138 KV	Reconductor line	\$1,195,000	8% (\$100,000)
WISSOTA – NSP 138 KV	Reconductor line	\$9,887,000	8% (\$826,000)
NSP – W MOUNT VERNON SUBSTATION 138 KV	Reconductor line	\$5,291,000	8% (\$442,000)
FERNWOOD – CHETNUT 138 KV	Reconductor line	\$2,863,000	8% (\$230,000)
FERNWOOD – BULAN 138 KV Line	Reconductor line	\$1,729,000	10% (\$182,000)
CONSOLIDATED COAL – CHESTNUT 138 KV L	Reconductor line	\$13,398,000	8% (\$1,127,000)
	Total allocated to project		\$60,869,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.